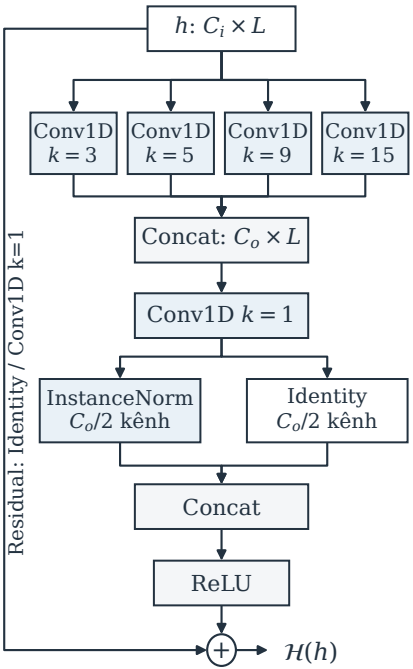
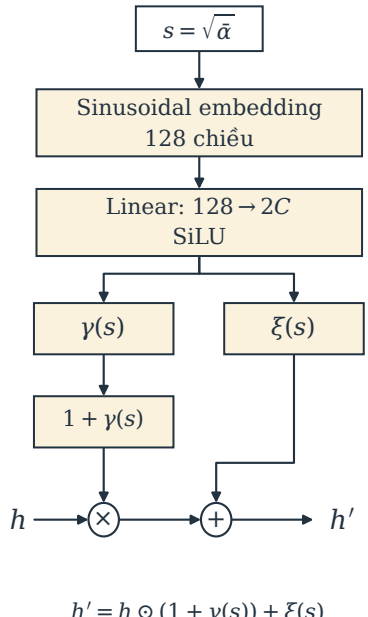
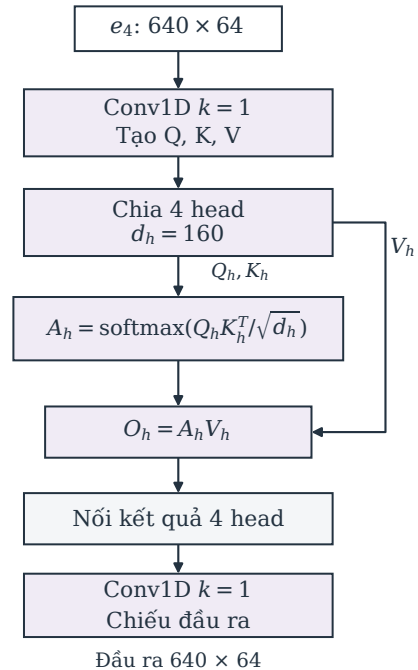

(a) Khối HNF
h: $C_i \times L$
Conv1D $k=3$
Conv1D $k=5$
Conv1D $k=9$
Conv1D $k=15$
Concat: $C_o \times L$
Conv1D $k=1$
InstanceNorm $C_o/2$ kênh
Identity $C_o/2$ kênh
Concat
ReLU
Residual: Identity / Conv1D k=1
$\mathcal{H}(h)$
Mỗi nhánh Conv1D sinh $C_o/4$ kênh.


(b) Điều chế FiLM
$s = \sqrt{\bar{\alpha}}$
Sinusoidal embedding 128 chiều
Linear: $128 \to 2C$ SiLU
$\gamma(s)$
$\xi(s)$
$1 + \gamma(s)$
h
h'
$h' = h \odot (1 + \gamma(s)) + \xi(s)$
Scale và shift theo kênh, broadcast trên chiều thời gian.


(c) Self-attention
e_4: 640×64
Conv1D $k=1$ Tạo Q, K, V
Chia 4 head $d_h = 160$
Q_h, K_h
V_h
$A_h = \text{softmax}(Q_h K_h^T / \sqrt{d_h})$
$O_h = A_h V_h$
Nối kết quả 4 head
Conv1D $k=1$ Chiếu đầu ra
Đầu ra 640 × 64
(kích thước minh họa với L = 512)